



Scientific evaluation of the chronic toxicity of the herbal medicine CGX in beagle dogs

Jang-Woo Shin^a, Hye-Jung Park^a, Min Kwon^b, Chang-Gue Son^{a,*}

^a Liver-Immune Research Center, Institute of Traditional Medicine and Bioscience of Daejeon University, 22-5, Daeheung-dong, Jung-gu, Daejeon 301-724, Republic of Korea

^b Safety Assessment Center, Korea Testing and Research Institute for Chemical Industry, 7-6, Gomak-ri, Wolgot-myeon, Gimpo-si, Kyonggi-do, Republic of Korea

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ABSTRACT

CGX is a potential hepatoprotective herbal medicine used to treat various chronic liver disorders. The purpose of the study was to evaluate the pharmaceutical safety of CGX via a systemic 13-week repeated dose toxicity test in beagle dogs. Male and female beagle dogs were divided into four groups and two animals each from the control and high-dose group (400 mg/kg) were allocated into recovery groups. The dogs were administered with CGX (0, 100, 200, 400 mg/kg) for 13 weeks. During the experimental period, the dogs were observed for signs of gross toxicity and for behavioral changes; body weight and food consumption were measured. An ophthalmologic examination and urinalysis were performed at 0 and 13th week and blood biochemistry and hematological parameters analyses were performed at 0, 6th, and 13th week. A histopathological examination was also performed at the end of the experiment. There were no CGX-induced abnormalities in clinical signs, organ weights, food consumption, hematological, urine, and blood biochemical parameters, or histopathological findings in any of the groups during or after the 13 weeks. We demonstrated the safety of CGX for 13-week repeated dose and considered that it is safe for chronic clinical use.

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1. Introduction

The use of herbal products as primary therapeutics or supplements for improving health-related conditions is popular worldwide (Seeff et al., 2001). The long history of clinical application and natural origin guarantee that herbal products are effective and non-toxic. Recently, concerns have been raised over the lack of quality control and of scientific evidence for the efficacy and safety of these products (Firenzuoli and Gori, 2007; Rousseau and Schachter, 2003). Several warnings have been issued regarding the potential adverse effects of herbal remedies, such as hepatotoxicity and nephrotoxicity (Wojcikowski et al., 2004; Stickel et al., 2005; Seeff, 2007; Tang et al., 2008).

In particular, safety evaluation of herbal drugs with multiple compositions is more difficult but intensively demanded. CGX is an herbal hepatotherapeutic consisting of 13 medicinal plants, developed from a traditional formula for the purpose of “liver cleaning”. CGX has been prescribed for patients with various chronic liver diseases including chronic viral B-type hepatitis, alcoholic disorder, and liver cirrhosis since 2001 (Cho et al.,

2001). Its hepatoprotective property previously were demonstrated in various animal models including alcoholic liver injury, acute liver injury, and liver fibrosis (Son et al., 2001; Shin et al., 2006; Hu et al., 2008).

A toxicity evaluation of CGX as a hepatotherapeutic regimen is very important if it is to be used on a large-scale. In an earlier study, the sub-acute toxicity of CGX was evaluated for 4 weeks, and the preparation received a “no-observable-adverse-effect-level” (NOAEL) rating (Choi et al., 2006). However, a chronic toxicity evaluation of CGX is necessary because most patients with chronic liver disease require long-term therapy.

We executed this systemic chronic toxicity in beagle dogs following a 13-week repeated CGX dose, and this result might be a classical example of chronic safety study for multi-herbal plant-derived medicine.

2. Materials and methods

2.1. Preparation of CGX

The CGX used in this study consisted of 5 g each of *Artemisia capillaris* herba, *Trionycis carapax*, and *Raphani semen*; 3 g each of *Atractylodis rhizoma alba*, *Poria cocos*, *Alismatis rhizoma*, *Atractylodis rhizoma*, and *Salvia miltiorrhizae* radix; 2 g each of *Polyporus*, *Amomi fructus*, and *Ponciri fructus*; and 1 g of *Glycyrrhizae radix* or *Aucklandiae radix*. All used herbs were satisfied with the Korean Pharmacopoeia (2007). CGX was manufactured in Samik Pharmacy (Seoul, Korea) approved Good

* Corresponding author. Address: Liver-Immune Research Center, Oriental Hospital of Daejeon University, 22-5 Daeheung-dong, Jung-gu, Daejeon 301-704, Republic of Korea. Tel.: +82 42 229 6723; fax: +82 42 257 6398.

E-mail address: ckson@dju.ac.kr (C.-G. Son).

Manufacturing Practice (GMP) by Korea Food and Drug Administration (KFDA). Briefly, 120 kg of the 13 herbs was boiled in 1200 l distilled water for 4 h at 100 °C and then filtered using a 300-mesh filter (50 µm). A 10.71% (w/w) lyophilized CGX aqueous extract was obtained from the dried mixture, and the tablet (1.2 g) containing 62% CGX (750 mg) was manufactured and administration dosages were prepared on the basis of CGX extracts.

Final CGX tablet were tested according to the Korean Pharmaceutical Codex (2007) and satisfied with criteria such as confirmation test for each herb, heavy metal (total heavy metal, <30 ppm; AS, <3 ppm; Pb, <5 ppm), general bacteria (<1 × 10⁵ CFU/g), fungi (<100 CFU/g), specific pathogens (negative for *Escherichia coli*, *Pseudomonas aeruginosa*, *Staphylococcus aureus*, and *Salmonella spp.*), tablet disintegration in water (<30 min), contained quality test (dimethylsuccinate, >0.15 mg/tablet; glycyrrizin, >1.1 mg/tablet).

Three dimensional chromatogram for CGX was analyzed using HPLC-DAD-MS for five major compositional herbs and their main compounds: Glycyrrhizae Radix, Artemisiae Capillaris Herba, Raphani Semen, Salvia Miltiorrhizae Radix, Atractylodis rhizoma alba and Ponciri Fructus vs liquiritin and glycyrrhizin, 6,7-dimethoxycoumarin and scopoletin, naringin, rosmarinic acid, atractylenolide III and poncirin, respectively. Briefly, after dissolution (30 mg of CGX tablet and 2 mg of five herbal extracts in 1 ml of water, 0.01 mg of seven standards in 1 ml of water or 50% methanol) and filtrated. The filtrates of these drugs were subjected to HPLC analysis. The HPLC system was consisted of SCL-10A system controller, LC-10AD pump, SPD-10MVP diode array detector and CTO-10AS column temperature controller (Shimadzu, Kyoto, Japan). A Phenomenex Prodigy C18 (2.0 × 150 mm) column was eluted with solvents A (10% acetonitrile in water containing 0.05% formic acid) and B (90% acetonitrile in water) at flow rate of 0.4 ml/min. Solutions 100% A and 0% B changing over 30 min to 25% B, 60 min to 75% B were used (Fig. 1).

2.2. Thirteen-week repeated dose toxicity test

2.2.1. Animals and management

Five-month-old male and female beagle dogs were obtained from Marshall Beijing (Beijing, China). The animal room was maintained at a temperature of 18.9–22.4 °C, with a relative humidity of 43.2–63.3%, air ventilation of 10–15 times/h, and ambient light (200–300 lux) controlled to produce a 12-h light/dark cycle. The dogs were acclimated to laboratory conditions for 3 weeks before use. The dogs were fed 320 ± 20 g per day Gold pet (Purina, Sungnam, Korea) twice per day and filtered water *ad libitum* during the study. Each dog was reared in a wire cage (900 × 1200 × 1000 mm) and identification cards stating the test and animal numbers were attached in the cages.

This study was performed at the Korea Testing and Research Institute following approval of the institutional Animal Care and Use Committee according to the "Guidelines for Toxicity Tests of Drugs and Related Materials, Document #2005-60" prepared by KFDA (2005).

2.2.2. Groups and administration

Sixteen male and 16 female dogs were divided into four groups of three animals each, then two dogs of each gender were allocated as control and a high-dose group, which served as a recovery group. CGX tablets containing 100, 200, and 400 mg/kg CGX extracts were orally administered once per day for 13 weeks in a gelatin capsule; a gelatin capsule only was administered to the controls. At the start of the experiment, the dogs weighed 6.418–8.140 kg (males) and 5.844–7.610 kg (female).

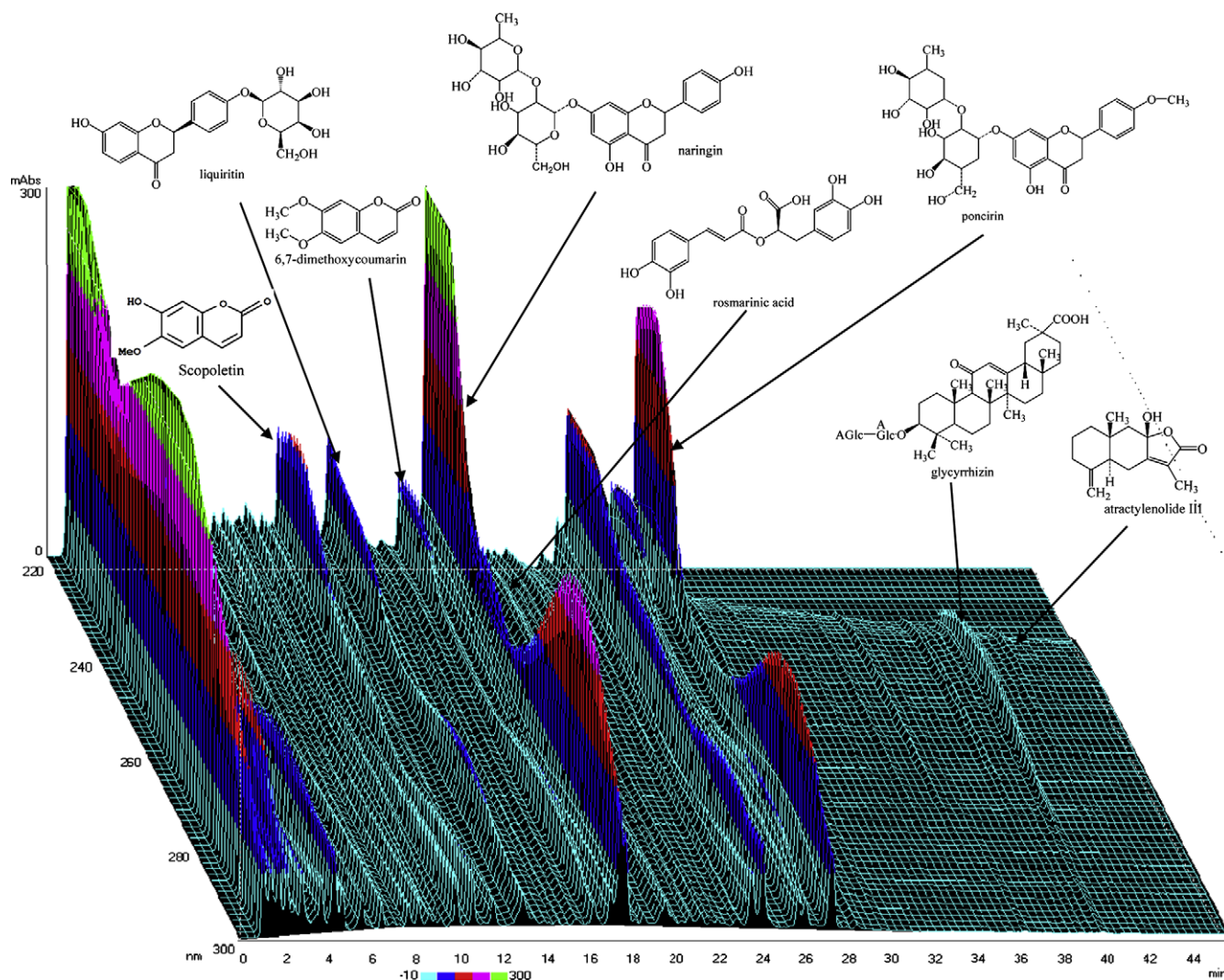


Fig. 1. Three-dimensional chromatogram of CGX by HPLC-DAD-MS and their major active compounds. CGX and eight standards were subjected to HPLC analysis. A Phenomenex Prodigy C18 (2.0 × 150 mm) column was eluted with solvents A (10% acetonitrile in water containing 0.05% formic acid) and B (90% acetonitrile in water) at flow rate of 0.4 ml/min.

2.2.3. Observation items and schedule

The dogs were observed for mortality, signs of gross toxicity, and behavioral changes at least once per day during the study. Body weight and food consumption were recorded every week. An ophthalmological examination was conducted by observing the appearance of the eyes with the naked eye before treatment and in the last week of the experiment. At the end of the experiment, all of the dogs were sacrificed after blood collection under deep anesthesia with Zoletil (Virbac Lab, Car-

ros, France). After exsanguination, internal organs (heart, liver, lungs, spleen, kidneys, adrenals, prostate, testes, epididymes, ovaries, brain, pituitary gland, uterus, submaxillary glands, and thyroids) were removed, and organ weights and gross findings were recorded. The testes and epididymes were fixed with Bouin's solution and the remaining organs including the tongue, thymus, duodenum, small intestine, large intestine, bone, trachea, urinary bladder, eyes, pancreas, spinal cord, vagina, seminal vesicles, esophagus, stomach, skin, and mesenteric lymph nodes were fixed with neutral buffered formalin and stored individually for histopathological examination. The dogs in the recovery groups were sacrificed after 2 weeks drug administration ceased, and all examinations were conducted as described above.

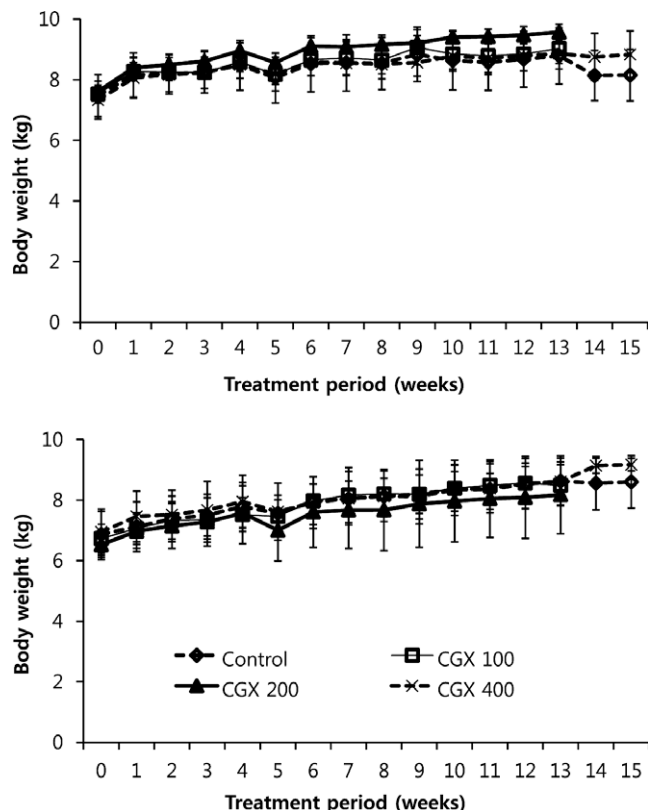


Fig. 2. Body weight changes in beagle dogs treated with CGX for 13 weeks. Sixteen male and female dogs were divided into four groups (three animals each) and an additional two dogs of each gender were allocated to a control and a high-dose group, which served as a recovery group. CGX (0, 100, 200, 400 mg/kg) was orally administered in a gelatin capsule once per day for 13 weeks.

2.2.4. Hematological, blood biochemical, and urinalysis

Blood was collected from the jugular vein after a 16-h fast on the day before the start of the experiment, at week 6, and at necropsy. Blood (2 ml) was transferred to a CBC bottle (EDTA 3K; Sewon Medical, Busan, Korea) for white blood cell count (WBC), red blood cell count (RBC), hemoglobin, hematocrit, mean corpuscular volume, mean corpuscular hemoglobin (MCH), mean corpuscular hemoglobin concentration (MCHC), and platelet analysis using an autohematoanalyzer (ADVIA120E; Bayer, New York, NY, USA). A portion of the blood was transferred to a Vacutainer (sodium citrate 3.2%; BD Biosciences, Franklin Lakes, NJ, USA) and centrifuged at 3000 rpm for 10 min. Plasma was isolated and used to determine the aggregation time with a coagulometer (Coagrex-100s; International Reagent Corporation, Hyogo, Japan).

Serum was isolated from the remaining blood and used for biochemical analyses with an automatic serum analyzer (model 7060; Hitachi, Tokyo, Japan) and Na/K/Cl analyzer (Bayer 644) to determine total protein (TP), albumin (ALB), albumin-globulin ratio (A/G), total bilirubin (T-BIL), aspartate aminotransferase (AST), alanine aminotransferase (ALT), alkaline phosphatase (ALP), creatinine (CREA), blood urea nitrogen (BUN), total cholesterol (CHOL), triglycerides (TG), glucose (GLU), calcium (CA), inorganic phosphorus (IP), creatinine kinase (CK), sodium (Na), potassium (K), and chloride (Cl), respectively.

Urine tests were performed before treatment and at necropsy using a Bayer Diagnostics Multistix 10SG strip and urine analyzer (Clinitek 500; Bayer) to determine specific gravity, pH, leucocytes, nitrite, protein, glucose, ketone, urobilinogen, bilirubin levels, and blood.

2.2.5. Histopathological examination

The organs of all groups were stained with hematoxylin–eosin (H&E) following fixation with 10% formalin or Bouin's solution (only for the testes and epididymes) and embedded in paraffin wax. A histopathologist performed a complete examination of the tissue samples.

2.2.6. Statistical analysis

The body weight, hematological data, blood biochemical data, and organ weights were analyzed for homogeneity of dispersion using Levene's test. One-way analysis of variance (ANOVA) was performed for data with significant dispersion, and this significance was confirmed with Dunnett's multiple inspection. All analyses were performed using SPSS program Ver. 10.1 (SPSS, Chicago, IL, USA).

Table 1

Final body weight and absolute organ weight of male beagle dogs treated with CGX (0, 100, 200, 400 mg/kg) for 13 weeks.

Dose (mg/kg/day)	Group				Recovery group	
	0 (n = 3)	100 (n = 3)	200 (n = 3)	400 (n = 3)	0 (n = 2)	400 (n = 2)
Body weight (kg)	8.898 ± 0.331	8.551 ± 0.475	9.095 ± 0.232	8.373 ± 0.544	7.815 ± 0.949	8.563 ± 0.856
Liver	242.63 ± 30.55	275.13 ± 25.08	272.57 ± 24.59	273.90 ± 31.34	272.85 ± 9.26	255.15 ± 11.38
Kidney (left)	18.63 ± 0.93	17.50 ± 2.75	19.47 ± 2.08	21.77 ± 0.81	21.00 ± 2.83	19.40 ± 0.00
Kidney (right)	18.40 ± 2.27	18.40 ± 2.86	19.73 ± 2.24	21.27 ± 2.56	20.45 ± 0.49	20.45 ± 4.03
Spleen	25.07 ± 3.96	19.40 ± 2.66	25.33 ± 4.57	19.90 ± 2.69	16.60 ± 2.97	27.30 ± 0.99
Adrenal gland (left)	0.54 ± 0.06	0.64 ± 0.06	0.44 ± 0.35	0.60 ± 0.06	0.68 ± 0.06	0.81 ± 0.12
Adrenal gland (right)	0.55 ± 0.05	0.56 ± 0.09	0.59 ± 0.07	0.60 ± 0.10	0.62 ± 0.06	0.81 ± 0.18
Testis (left)	5.44 ± 0.54	6.04 ± 0.41	5.48 ± 0.52	6.71 ± 0.50	4.98 ± 1.63	6.53 ± 0.72
Testis (right)	5.63 ± 0.51	7.41 ± 2.23	5.09 ± 1.01	6.23 ± 0.57	5.43 ± 1.59	5.53 ± 2.47
Brain	74.20 ± 4.33	71.77 ± 4.11	70.60 ± 2.10	71.03 ± 3.56	73.65 ± 4.17	76.80 ± 0.14
Pituitary gland	0.06 ± 0.03	0.06 ± 0.05	0.08 ± 0.02	0.05 ± 0.01	0.04 ± 0.00	0.17 ± 0.20
Lung	85.00 ± 22.98	71.50 ± 6.08	96.90 ± 8.35	82.27 ± 7.22	76.30 ± 2.26	85.00 ± 18.95
Heart	75.37 ± 1.98	75.07 ± 10.65	74.17 ± 5.32	69.57 ± 5.83	67.00 ± 0.14	80.20 ± 7.92
Submaxillary gland	7.69 ± 0.10	8.82 ± 0.37	8.84 ± 0.65	8.27 ± 0.47	9.85 ± 0.92	10.55 ± 1.86
Prostate gland	0.38 ± 0.08	0.35 ± 0.07	0.34 ± 0.09	0.37 ± 0.02	0.40 ± 0.01	0.55 ± 0.15
Thyroid gland (left)	0.43 ± 0.08	0.035 ± 0.05	0.33 ± 0.02	0.37 ± 0.06	0.31 ± 0.04	0.35 ± 0.06
Thyroid gland (right)	3.78 ± 0.63	2.38 ± 1.58	2.75 ± 1.00	2.78 ± 1.24	1.63 ± 0.76	2.99 ± 0.47
Epididymis (left)	1.09 ± 0.16	1.01 ± 0.23	0.90 ± 0.10	1.09 ± 0.16	0.84 ± 0.13	1.14 ± 0.23
Epididymis (right)	1.14 ± 0.20	1.08 ± 0.28	0.84 ± 0.09	1.09 ± 0.03	0.98 ± 0.05	1.19 ± 0.30

Values represent means (g) ± SD, n = number of animals.

Table 2

Relative organ weight of male beagle dogs treated with CGX (0, 100, 200, 400 mg/kg) for 13 weeks.

Dose (mg/kg/day)	Group				Recovery group	
	0 (n = 3)	100 (n = 3)	200 (n = 3)	400 (n = 3)	0 (n = 2)	400 (n = 2)
Liver	2.734 ± 0.405	3.227 ± 0.390	2.999 ± 0.293	3.297 ± 0.607	3.525 ± 0.546	3.001 ± 0.433
Kidney (left)	0.210 ± 0.012	0.206 ± 0.041	0.215 ± 0.028	0.261 ± 0.027	0.268 ± 0.004	0.228 ± 0.023
Kidney (right)	0.207 ± 0.029	0.216 ± 0.043	0.217 ± 0.030	0.256 ± 0.045	0.263 ± 0.026	0.238 ± 0.023
Spleen	0.281 ± 0.034	0.228 ± 0.037	0.278 ± 0.045	0.237 ± 0.021	0.212 ± 0.012	0.320 ± 0.020
Adrenal gland (left)	0.006 ± 0.001	0.008 ± 0.001	0.005 ± 0.004	0.007 ± 0.001	0.009 ± 0.002	0.009 ± 0.000
Adrenal gland (right)	0.006 ± 0.001	0.007 ± 0.001	0.006 ± 0.001	0.007 ± 0.002	0.008 ± 0.002	0.009 ± 0.001
Testis (left)	0.061 ± 0.008	0.071 ± 0.004	0.060 ± 0.006	0.080 ± 0.002	0.063 ± 0.013	0.076 ± 0.001
Testis (right)	0.063 ± 0.004	0.086 ± 0.022	0.056 ± 0.012	0.074 ± 0.005	0.069 ± 0.012	0.063 ± 0.023
Brain	0.834 ± 0.025	0.842 ± 0.083	0.777 ± 0.037	0.853 ± 0.101	0.953 ± 0.169	0.901 ± 0.092
Pituitary gland	0.001 ± 0.000	0.001 ± 0.001	0.001 ± 0.000	0.001 ± 0.000	0.001 ± 0.000	0.002 ± 0.002
Lung	0.962 ± 0.298	0.838 ± 0.090	1.065 ± 0.078	0.987 ± 0.132	0.982 ± 0.090	0.987 ± 0.123
Heart	0.848 ± 0.054	0.877 ± 0.101	0.815 ± 0.047	0.836 ± 0.128	0.864 ± 0.103	0.937 ± 0.001
Submaxillary gland	0.086 ± 0.002	0.103 ± 0.007	0.097 ± 0.009	0.099 ± 0.012	0.126 ± 0.004	0.123 ± 0.009
Prostate gland	0.004 ± 0.001	0.004 ± 0.001	0.004 ± 0.001	0.004 ± 0.000	0.005 ± 0.001	0.006 ± 0.002
Thyroid gland (left)	0.005 ± 0.001	0.004 ± 0.001	0.004 ± 0.000	0.004 ± 0.001	0.004 ± 0.000	0.004 ± 0.001
Thyroid gland (right)	0.043 ± 0.008	0.019 ± 0.022	0.030 ± 0.012	0.033 ± 0.013	0.020 ± 0.007	0.035 ± 0.002
Epididymis (left)	0.012 ± 0.002	0.012 ± 0.003	0.010 ± 0.001	0.013 ± 0.001	0.011 ± 0.000	0.013 ± 0.001
Epididymis (right)	0.013 ± 0.002	0.013 ± 0.003	0.009 ± 0.001	0.013 ± 0.001	0.013 ± 0.001	0.014 ± 0.002

Values represent means (g) ± SD, n = number of animals.

* Significantly different from control group at $P < 0.05$ with Scheffe test.**Table 3**

Final body weight and absolute organ weight of female dogs treated with CGX (0, 100, 200, 400 mg/kg) for 13 weeks.

Dose (mg/kg/day)	Group				Recovery group	
	0 (n = 3)	100 (n = 3)	200 (n = 3)	400 (n = 3)	0 (n = 2)	400 (n = 2)
Body weight	8.174 ± 0.589	8.113 ± 0.553	7.691 ± 1.317	7.785 ± 0.971	8.379 ± 0.921	8.890 ± 0.238
Liver	220.97 ± 23.45	225.07 ± 26.83	246.47 ± 5.77	246.07 ± 27.00	219.40 ± 37.76	255.20 ± 32.24
Kidney (left)	16.33 ± 1.46	16.73 ± 1.27	17.27 ± 1.66	16.06 ± 1.34	15.85 ± 3.32	18.00 ± 0.71
Kidney (right)	16.27 ± 0.74	16.33 ± 1.21	16.60 ± 2.52	16.07 ± 1.29	15.00 ± 4.24	18.20 ± 1.84
Spleen	22.27 ± 2.78	24.47 ± 3.01	23.53 ± 6.21	26.09 ± 2.67	25.50 ± 10.47	30.60 ± 0.42
Adrenal gland (left)	0.56 ± 0.03	0.67 ± 0.09	0.51 ± 0.13	0.55 ± 0.10	0.56 ± 0.06	0.64 ± 0.22
Adrenal gland (right)	0.59 ± 0.04	0.67 ± 0.22	0.54 ± 0.09	0.51 ± 0.12	0.71 ± 0.11	0.65 ± 0.18
Ovary (left)	0.37 ± 0.18	0.39 ± 0.15	0.33 ± 0.07	0.25 ± 0.06	0.23 ± 0.19	0.30 ± 0.14
Ovary (right)	0.56 ± 0.51	0.36 ± 0.04	0.33 ± 0.07	0.27 ± 50.03	0.23 ± 0.06	0.33 ± 0.01
Brain	57.48 ± 24.92	67.30 ± 1.47	67.47 ± 1.37	67.57 ± 4.09	68.65 ± 0.21	70.70 ± 4.67
Pituitary gland	0.03 ± 0.02	0.06 ± 0.01	0.05 ± 0.03	0.05 ± 0.01	0.07 ± 0.02	0.05 ± 0.01
Lung	95.77 ± 24.55	81.80 ± 6.88	69.13 ± 12.62	72.40 ± 8.92	68.65 ± 1.34	92.45 ± 12.52
Heart	65.15 ± 8.61	67.43 ± 3.35	64.50 ± 9.56	70.07 ± 9.00	64.00 ± 5.66	70.85 ± 0.35
Submaxillary gland	7.85 ± 0.52	6.75 ± 0.57	7.61 ± 1.03	5.95 ± 1.08	7.11 ± 0.74	7.57 ± 0.45
Thyroid gland (left)	0.32 ± 0.02	0.34 ± 0.12	0.47 ± 0.07	0.30 ± 0.05	0.43 ± 0.01	0.35 ± 0.06
Thyroid gland (right)	0.30 ± 0.02	0.29 ± 0.08	0.40 ± 0.07	0.27 ± 0.06	0.46 ± 0.05	0.44 ± 0.06
Uterus	11.89 ± 15.36	13.68 ± 19.22	3.88 ± 2.52	4.51 ± 0.34	2.30 ± 0.66	8.42 ± 0.00

Values represent means (g) ± SD, n = number of animals.

Table 4

Relative organ weight of female dogs treated with CGX (0, 100, 200, 400 mg/kg) for 13 weeks.

Dose (mg/kg/day)	Group				Recovery group	
	0 (n = 3)	100 (n = 3)	200 (n = 3)	400 (n = 3)	0 (n = 2)	400 (n = 2)
Liver	2.699 ± 0.091	2.779 ± 0.325	3.270 ± 0.576	3.165 ± 0.094	2.542 ± 0.184	2.788 ± 0.415
Kidney (left)	0.200 ± 0.006	0.206 ± 0.007	0.229 ± 0.048	0.207 ± 0.011	0.188 ± 0.019	0.203 ± 0.013
Kidney (right)	0.199 ± 0.005	0.201 ± 0.009	0.219 ± 0.044	0.208 ± 0.027	0.177 ± 0.031	0.205 ± 0.015
Spleen	0.272 ± 0.025	0.301 ± 0.021	0.304 ± 0.045	0.340 ± 0.066	0.299 ± 0.092	0.344 ± 0.014
Adrenal gland (left)	0.007 ± 0.001	0.008 ± 0.002	0.007 ± 0.001	0.007 ± 0.002	0.007 ± 0.000	0.007 ± 0.003
Adrenal gland (right)	0.007 ± 0.001	0.008 ± 0.003	0.007 ± 0.001	0.007 ± 0.002	0.008 ± 0.000	0.007 ± 0.002
Ovary (left)	0.005 ± 0.002	0.005 ± 0.002	0.004 ± 0.000	0.003 ± 0.001	0.003 ± 0.002	0.003 ± 0.002
Ovary (right)	0.007 ± 0.007	0.004 ± 0.000	0.004 ± 0.000	0.004 ± 0.001	0.003 ± 0.000	0.004 ± 0.000
Brain	0.700 ± 0.298	0.833 ± 0.072	0.896 ± 0.170	0.875 ± 0.106	0.824 ± 0.093	0.796 ± 0.074
Pituitary gland	0.000 ± 0.000	0.001 ± 0.000	0.001 ± 0.000	0.001 ± 0.000	0.001 ± 0.000	0.001 ± 0.000
Lung	0.897 ± 0.618	1.015 ± 0.158	0.898 ± 0.013	0.947 ± 0.221	0.823 ± 0.074	1.038 ± 0.113
Heart	0.795 ± 0.048	0.834 ± 0.067	0.845 ± 0.091	0.906 ± 0.139	0.765 ± 0.017	0.797 ± 0.017
Submaxillary gland	0.096 ± 0.010	0.083 ± 0.010	0.100 ± 0.014	0.076 ± 0.008	0.085 ± 0.000	0.085 ± 0.007
Thyroid gland (left)	0.004 ± 0.000	0.004 ± 0.001	0.006 ± 0.001	0.004 ± 0.001	0.005 ± 0.000	0.004 ± 0.001
Thyroid gland (right)	0.004 ± 0.000	0.004 ± 0.001	0.005 ± 0.001	0.004 ± 0.001	0.005 ± 0.000	0.005 ± 0.001
Uterus	0.152 ± 0.199	0.162 ± 0.223	0.049 ± 0.026	0.059 ± 0.011	0.028 ± 0.011	0.095 ± 0.003

Values represent means (g) ± SD, n = number of animals.

Table 5

Blood biochemical parameters of male dogs treated with CGX (0, 100, 200, 400 mg/kg) for 13 weeks.

Dose (mg/kg/day)	0 week				6 weeks				13 weeks				Recovery	
	0 (n = 5)	100 (n = 3)	200 (n = 3)	400 (n = 5)	0 (n = 5)	100 (n = 3)	200 (n = 3)	400 (n = 5)	0 (n = 3)	100 (n = 3)	200 (n = 3)	400 (n = 3)	0 (n = 2)	400 (n = 2)
TP (g/dl)	5.0 ± 0.1	5.1 ± 0.2	5.2 ± 0.4	5.1 ± 0.3	5.1 ± 0.1	5.2 ± 0.3	5.2 ± 0.1	5.1 ± 0.3	5.0 ± 0.1	5.4 ± 0.4	5.3 ± 0.2	5.2 ± 0.1	5.4 ± 0.3	5.4 ± 0.3
ALB (g/dl)	2.4 ± 0.1	2.5 ± 0.2	2.5 ± 0.2	2.4 ± 0.1	2.4 ± 0.2	2.6 ± 0.2	2.5 ± 0.2	2.4 ± 0.1	2.4 ± 0.1	2.8 ± 0.2	2.6 ± 0.1	2.5 ± 0.1	2.5 ± 0.0	2.7 ± 0.1
A/G	0.9 ± 0.1	0.9 ± 0.1	0.9 ± 0.1	0.9 ± 0.0	0.9 ± 0.1	1.0 ± 0.0	0.9 ± 0.1	0.9 ± 0.1	1.0 ± 0.1	1.1 ± 0.1	0.9 ± 0.1	0.9 ± 0.1	0.9 ± 0.1	1.0 ± 0.1
T-BIL (mg/dl)	0.07 ± 0.01	0.08 ± 0.01	0.07 ± 0.02	0.06 ± 0.02	0.05 ± 0.02	0.05 ± 0.04	0.05 ± 0.01	0.06 ± 0.02	0.07 ± 0.01	0.08 ± 0.02	0.09 ± 0.01	0.08 ± 0.01	0.08 ± 0.01	0.44 ± 0.51
ALP (U/l)	348 ± 74	304 ± 50	299 ± 31	311 ± 70	405 ± 61	317 ± 37	335 ± 15	338 ± 67	285 ± 80	267 ± 18	276 ± 27	276 ± 81	365 ± 106	235 ± 43
AST (U/l)	34 ± 5	33 ± 8	29 ± 10	31 ± 9	31 ± 3	29 ± 5	30 ± 6	32 ± 6	32 ± 2	32 ± 4	32 ± 5	29 ± 3	31 ± 2	35 ± 5
ALT (U/l)	30 ± 8	26 ± 3	36 ± 18	29 ± 3	33 ± 8	30 ± 6	30 ± 6	30 ± 3	32 ± 10	29 ± 6	30 ± 4	27 ± 3	34 ± 10	36 ± 1
CREA (mg/dl)	0.7 ± 0.1	0.7 ± 0.1	0.7 ± 0.0	0.7 ± 0.0	0.7 ± 0.0	0.7 ± 0.1	0.7 ± 0.0	0.7 ± 0.1	0.8 ± 0.0	0.7 ± 0.1	0.9 ± 0.1	0.7 ± 0.0	0.8 ± 0.1	0.8 ± 0.0
BUN (mg/dl)	6.7 ± 0.8	7.6 ± 2.8	6.9 ± 0.4	7.3 ± 1.4	7.9 ± 1.7	7.0 ± 0.7	7.8 ± 0.6	8.1 ± 1.4	8.6 ± 2.5	8.9 ± 1.3	9.4 ± 0.8	10.2 ± 1.2	9.4 ± 2.9	10.4 ± 2.5
CHOL (mg/dl)	133 ± 18	128 ± 35	167 ± 32	172 ± 20	147 ± 12	162 ± 19	168 ± 27	168 ± 14	145 ± 26	170 ± 27	189 ± 43	193 ± 15	173 ± 35	187 ± 1
TG (mg/dl)	24 ± 5	26 ± 11	27 ± 13	32 ± 7	34 ± 12	30 ± 9	29 ± 9	26 ± 8	30 ± 14	40 ± 14	34 ± 8	30 ± 16	42 ± 21	36 ± 17
GLU (mg/dl)	96 ± 4	92 ± 14	95 ± 5	101 ± 9	97 ± 3	103 ± 10	101 ± 3	98 ± 8	91 ± 7	92 ± 11	94 ± 3	87 ± 3	95 ± 9	102 ± 8
CA (mg/dl)	10.7 ± 0.1	10.9 ± 0.5	11.0 ± 0.1	11.0 ± 0.3	10.6 ± 0.4	10.7 ± 0.3	10.8 ± 0.3	10.5 ± 0.2	10.3 ± 0.3	10.6 ± 0.2	10.4 ± 0.1	10.1 ± 0.2	10.2 ± 0.6	10.3 ± 0.6
IP (mg/dl)	7.3 ± 0.4	7.6 ± 0.4	7.6 ± 0.3	7.5 ± 0.6	6.4 ± 0.4	6.0 ± 0.3	6.4 ± 0.4	6.1 ± 0.7	5.9 ± 0.6	5.7 ± 0.3	5.8 ± 0.4	5.8 ± 0.7	5.4 ± 0.4	5.6 ± 0.1
GGT (U/l)	4.3 ± 1.6	4.7 ± 2.1	5.2 ± 0.7	4.8 ± 1.7	3.8 ± 0.5	3.0 ± 1.6	3.6 ± 1.4	3.3 ± 0.4	4.2 ± 0.7	3.1 ± 0.7	4.5 ± 1.3	3.7 ± 0.6	4.6 ± 0.1	4.4 ± 0.6
LDH (IU/l)	267 ± 192	335 ± 383	153 ± 115	206 ± 200	64 ± 22	63 ± 12	47 ± 3	57 ± 17	84 ± 15	114 ± 49	73 ± 30	84 ± 32	56 ± 20	63 ± 30
CK (IU/l)	299 ± 59	296 ± 181	268 ± 92	246 ± 122	175 ± 14	156 ± 22	183 ± 36	207 ± 60	196 ± 34	207 ± 46	209 ± 13	180 ± 64	132 ± 23	178 ± 7
Na (mmol/l)	145 ± 1	144 ± 1	144 ± 1	144 ± 2	146 ± 1	147 ± 1	148 ± 1	147 ± 1	146 ± 1	148 ± 1	147 ± 2	145 ± 1	145 ± 1	147 ± 3
K (mmol/l)	5.09 ± 0.32	5.24 ± 0.01	4.93 ± 0.19	4.89 ± 0.22	4.87 ± 0.20	4.64 ± 0.32	4.47 ± 0.31	4.60 ± 0.14	4.79 ± 0.37	4.87 ± 0.44	4.56 ± 0.02	4.57 ± 0.17	4.40 ± 0.35	4.27 ± 0.08
Cl (mmol/l)	110 ± 1	109 ± 2	111 ± 1	111 ± 2	112 ± 3	111 ± 2	112 ± 2	112 ± 1	113 ± 1	114 ± 1	116 ± 2	116 ± 1	111 ± 1	113 ± 0

Values represent means (g) ± SD, n = number of animals.

* Significantly different from control group at $P < 0.05$ with Scheffe test. See text for abbreviations.**Table 6**

Blood biochemical parameters of female dogs treated with CGX (0, 100, 200, 400 mg/kg) for 13 weeks.

Dose (mg/kg/day)	0 week				6 weeks				13 weeks				Recovery	
	0 (n = 5)	100 (n = 3)	200 (n = 3)	400 (n = 5)	0 (n = 5)	100 (n = 3)	200 (n = 3)	400 (n = 5)	0 (n = 3)	100 (n = 3)	200 (n = 3)	400 (n = 3)	0 (n = 2)	400 (n = 2)
TP (g/dl)	5.1 ± 0.3	5.2 ± 0.1	5.0 ± 0.1	5.0 ± 0.1	5.3 ± 0.3	5.3 ± 0.3	5.2 ± 0.2	5.5 ± 0.2	5.3 ± 0.2	5.3 ± 0.2	5.3 ± 0.2	5.4 ± 0.1	5.4 ± 0.1	5.2 ± 0.1
ALB (g/dl)	2.5 ± 0.1	2.6 ± 0.1	2.3 ± 0.2	2.4 ± 0.1	2.6 ± 0.1	2.6 ± 0.1	2.5 ± 0.1	2.7 ± 0.1	2.7 ± 0.1	2.7 ± 0.1	2.6 ± 0.1	2.8 ± 0.1	2.7 ± 0.2	2.7 ± 0.0
A/G	1.0 ± 0.1	1.0 ± 0.0	0.9 ± 0.1	0.9 ± 0.1	1.0 ± 0.1	1.0 ± 0.1	0.9 ± 0.1	1.0 ± 0.1	1.0 ± 0.1	1.0 ± 0.1	1.0 ± 0.1	1.1 ± 0.1	1.0 ± 0.1	1.1 ± 0.0
T-BIL (mg/dl)	0.07 ± 0.02	0.06 ± 0.01	0.06 ± 0.01	0.07 ± 0.01	0.08 ± 0.04	0.08 ± 0.03	0.07 ± 0.03	0.08 ± 0.02	0.13 ± 0.04	0.10 ± 0.04	0.11 ± 0.06	0.09 ± 0.03	0.07 ± 0.03	0.07 ± 0.00
ALP (U/l)	319 ± 50	418 ± 143	301 ± 55	331 ± 64	393 ± 104	447 ± 219	332 ± 28	391 ± 67	315 ± 73	321 ± 160	282 ± 59	310 ± 41	298 ± 109	307 ± 107
AST (U/l)	42 ± 25	45 ± 27	30 ± 5	32 ± 8	38 ± 7	33 ± 7	37 ± 9	36 ± 11	44 ± 16	31 ± 7	38 ± 7	40 ± 10	37 ± 5	33 ± 0
ALT (U/l)	123 ± 184	147 ± 203	29 ± 5	2 ± 8	47 ± 19	40 ± 22	27 ± 7	30 ± 7	50 ± 22	42 ± 22	36 ± 15	35 ± 6	36 ± 16	29 ± 10
CREA (mg/dl)	0.8 ± 0.1	0.7 ± 0.1	0.7 ± 0.1	0.7 ± 0.1	0.8 ± 0.1	0.7 ± 0.1	0.7 ± 0.1	0.7 ± 0.0	0.9 ± 0.1	0.8 ± 0.1	0.8 ± 0.1	0.8 ± 0.0	0.9 ± 0.1	0.8 ± 0.1
BUN (mg/dl)	9.7 ± 1.9	9.0 ± 3.1	7.1 ± 0.7	7.8 ± 1.5	11.9 ± 3.1	9.8 ± 1.9	9.4 ± 2.3	9.6 ± 0.9	13.2 ± 3.4	9.9 ± 3.3	10.9 ± 1.8	11.0 ± 2.0	12.7 ± 2.4	9.4 ± 1.6
CHOL (mg/dl)	147 ± 21	159 ± 34	141 ± 11	144 ± 14	151 ± 20	158 ± 56	147 ± 10	158 ± 15	169 ± 13	167 ± 49	163 ± 8	186 ± 6	152 ± 1	159 ± 28
TG (mg/dl)	24 ± 4	29 ± 14	25 ± 9	24 ± 8	31 ± 7	29 ± 4	28 ± 1	23 ± 6	27 ± 10	31 ± 2	38 ± 9	27 ± 6	38 ± 20	27 ± 9
GLU (mg/dl)	95 ± 9	105 ± 15	91 ± 5	92 ± 11	96 ± 18	103 ± 9	93 ± 4	95 ± 11	97 ± 12	100 ± 9	88 ± 5	99 ± 12	109 ± 22	86 ± 7
CA (mg/dl)	10.8 ± 0.4	10.7 ± 0.3	10.8 ± 0.3	10.6 ± 0.2	10.9 ± 0.6	11.0 ± 0.2	10.6 ± 0.5	10.8 ± 0.3	10.4 ± 0.4	10.4 ± 0.5	10.3 ± 0.1	10.3 ± 0.5	10.5 ± 0.1	10.2 ± 0.2
IP (mg/dl)	7.0 ± 0.5	7.2 ± 0.3	7.0 ± 0.3	7.1 ± 0.4	5.2 ± 0.7	5.7 ± 0.5	5.8 ± 0.3	5.6 ± 0.8	3.7 ± 1.1	4.9 ± 0.2	5.1 ± 0.3	4.5 ± 1.0	6.1 ± 0.4	5.5 ± 0.4
GGT (U/l)	4.2 ± 0.3	4.5 ± 1.2	4.4 ± 1.5	4.1 ± 0.9	2.8 ± 0.2	3.3 ± 0.4	3.4 ± 1.0	3.0 ± 0.5	3.8 ± 0.6	4.0 ± 0.8	3.9 ± 1.8	2.9 ± 0.9	3.0 ± 1.1	3.2 ± 0.2
LDH (IU/l)	17 ± 44	164 ± 56	122 ± 41	211 ± 50	80 ± 23	67 ± 11	72 ± 10	72 ± 26	87 ± 35	68 ± 7	79 ± 22	60 ± 19	92 ± 23	74 ± 20
CK (IU/l)	200 ± 24	199 ± 47	171 ± 39	216 ± 49	282 ± 105	207 ± 60	266 ± 143	232 ± 90	396 ± 278	186 ± 49	236 ± 96	232 ± 88	197 ± 6	182 ± 8
Na (mmol/l)	144 ± 1	145 ± 1	144 ± 0	144 ± 2	150 ± 2	149 ± 2	150 ± 2	150 ± 1	147 ± 1	146 ± 2	148 ± 3	147 ± 1	146 ± 1	145 ± 1
K (mmol/l)	4.89 ± 0.15	4.63 ± 0.14	4.95 ± 0.07	4.92 ± 0.24	4.36 ± 0.27	4.34 ± 0.17	4.58 ± 0.20	4.64 ± 0.32	4.06 ± 0.36	4.15 ± 0.07	4.57 ± 0.06	4.19 ± 0.10	4.70 ± 0.25	4.61 ± 0.23
Cl (mmol/l)	111 ± 1	110 ± 1	112 ± 1	111 ± 1	112 ± 1	111 ± 1	113 ± 0	113 ± 1	117 ± 3	116 ± 2	116 ± 3	116 ± 2	112 ± 1	112 ± 1

Values represent means (g) ± SD, n = number of animals. See text for abbreviations.

Table 7

Histopathological findings in beagle dogs treated with CGX (0, 100, 200, 400 mg/kg) for 13 weeks.

	Histopathological finding	Sex	Male						Female						
			Dose (mg/kg)				Recovery		Dose (mg/kg)				Recovery		
			0	100	200	400	0	400	0	100	200	400	0	400	
			3	3	3	3	2	2	3	3	3	3	2	2	
		No. of animals													
Liver	Submucosal inflammatory cells aggregation/multifocal/minimal	1	1												
	Submucosal inflammatory cells aggregation/focal/minimal	1	1			1									
	Submucosal inflammatory cells/focal/minimal										2	1	1		
	Submucosal inflammatory cells infiltration									1				1	
Kidney (Left)	Inflammatory cells infiltration/multifocal/minimal			1											
	Interstitial inflammatory cells/focal/minimal	1													
	Papillary mineralization/multifocal/minimal	1	2	2	2		1	2	3	2	2	1	1		
	Pelvic mineralization/focal/minimal					1									
Kidney (Right)	Medullary rays–vacuolation/multifocal/minimal	1	1	1				1					1		
	Pelvic mineralization/multifocal/minimal		1	1	2	1	1	1			1		1		
	Medullary rays–vacuolation/multifocal/minimal							1					1		
	Cyst in pars distalis				1				1	1			2		
Pituitary gland	Interstitial inflammatory cells infiltration		1					–	–	–	–	–	–	–	
Prostate	Granulomatous inflammatory cells/focal	1													
Lung	Perivascular/multifocal			2				1		1			1		
	Peribronchiolar inflammatory cells/multifocal														
	Present			1				2					1		
	Minimal			1											
	Moderate			1						1					
	Interstitial inflammatory cells/multifocal/minimal				2										
Testis (Left)	Weak immature			1				–	–	–	–	–	–	–	
Testis (Right)	Weak immature			1				–	–	–	–	–	–	–	
Ovary (Left)	No corpus luteum	–	–	–	–	–	–							1	
Spleen	Agonal congestion	2		1			1	1	1	1					

3. Results

3.1. Survival and clinical signs

No animals died in any of the groups during the experimental period (13 weeks). There were no abnormal clinical signs in any of the groups, and food consumption was not significantly different between the CGX-treated and control groups (data not shown).

3.2. Body weight and organ weights

The changes in the body weight of the CGX-treated groups were not significantly different from those of the control group during the experimental period (Fig. 2). No significant differences in absolute and relative organ weights were found among the four groups, except for testis (Tables 1–4). The relative weight of the left testis in the high dose male group was significantly higher than in the control group ($P < 0.05$). However, there was no difference in the right testis or in absolute organ weight.

3.3. Hematological, blood biochemical and urine parameters

There were no significant changes in hematological parameters in the CGX-treated group (data not shown). Chloride level increased significantly in the male high-dose group compared to the control group ($P < 0.05$). However, the change was within the physiological range, and not significant in females (Tables 5 and 6). Other blood chemical parameters were not changed.

No significant differences were found between the CGX-treated groups and the controls for specific gravity, pH, leukocyte, nitrite, protein, glucose, ketone, urobilinogen, bilirubin, and blood in the urine (data not shown).

3.4. Ophthalmologic and histopathological findings

There were no abnormal ophthalmologic findings in any of the groups (data not shown). The major histopathological findings

were focal or multifocal infiltration and aggregation of submucosal inflammatory cells in the liver, papillary mineralization in the kidney, and agonal congestion in the spleen (Table 7); however, these findings were consistently noted in control dogs. There were no significant findings in any of the CGX-treated groups compared to the control group.

4. Discussion

A variety of traditional remedies or herbal-derived compounds have been investigated for the treatment of hepatic disorders. Approximately 20% of patients with hepatic diseases use wide variety of herbal products (Bent, 2008). Herbal based active compounds such as Silymarin, glycyrrhizin, and medicinal herbs like *Phyllanthus amarus*, *Picrorrhiza kurroa*, and liver support formula LIV 52, sho-saiko-to have been prescribed for patients suffering with hepatic disorders (Levy et al., 2004; Stickel and Schuppan, 2007). CGX was developed as a hepatoprotective drug from a traditional herbal formula and is prescribed primarily for patients with chronic viral hepatitis or alcoholic disease (Cho et al., 2001; Shin et al., 2006).

Following our previously study of 4-week repeated toxicity test for CGX (Choi et al., 2006) here, we evaluated the chronic toxicity of CGX via a systemic 13-week repeated dose toxicity study. The therapeutic dosage of CGX for clinical application is 60–100 mg/kg per day, so we treated both male and female beagle dogs with 100, 200 or 400 mg/kg/day CGX for 13 weeks. No abnormal clinical signs, including changes in food consumption and body weight, were observed in any of the groups. Because the adverse signs of most drugs include changes in appetite or in feces, these results indicated that CGX was not initially toxic.

The relative weight of the left testis in the high dose administered male dogs was increased. However, this change was not observed in right testis weight in any groups and no dose dependent correlation was observed. Further, no abnormal effects were seen in histopathological evaluation of testis tissue. Therefore, this change was not considered as CGX-induced abnormality.

There were no significant changes in hematological or blood biochemical parameters in any of the treated groups, except for a significant increase in chloride level; the increase, however, was within the physiological range, and occurred only in the male high-dose group. There was no abnormal urine or ophthalmologic findings. Histopathological studies showed minimal focal or multifocal infiltration of inflammatory cells in the liver and kidney and minimal papillary mineralization in the kidney of CGX-treated groups as well as in the controls. Therefore, these results lead to the anticipation that minimal histopathological changes may not be due to the administration of CGX formulation during experimental period.

In general, the present medical therapeutics depend on scientific knowledge to improve health. However, still there are certain aspects of non-scientific approach in medicinal practices. Among them medicinal foods or herbal formulations are often not subjected to toxicity testing before prescribing to humans. As the use of herbal products has increased worldwide, the safety of herbal remedies has become an important issue (Seeff, 2007; Tang et al., 2008). Many of the adverse effects of herbal medicines are related to plant misidentification, incorrect preparation, heavy metal contamination, or adulteration (Ernst, 2004). In addition, the interaction of some herbal ingredients and Western medicines are also an issue (Hu et al., 2005; Dasgupta and Bernard, 2006). Liver toxicity and its related disorders are considered as the most common concern, because the liver is a prime drug target (Williams and Iatropoulos, 2002). Recent reports have noted possible deleterious effects of sho-saiko-to and LIV 52 in patients with liver diseases (Itoh et al., 1995; Ikegami et al., 2003; Levy et al., 2004; Hsu et al., 2006).

The herbal product CGX was developed based on the long history of clinical application of the traditional remedy and after functional scientific studies. This study found that the NOAEL of CGX was greater than 400 mg/kg/day, which represents four times than that of recommended dose for clinical applications. The results of this study strongly suggest that CGX could be safe for chronic clinical use, although differences in drug metabolism among species and the possibility of an interaction with other drugs may have an effect (Shayne, 2007). The safety of CGX in the management of liver dysfunctions in human should be monitored carefully in a further study.

Conflict of Interest

The authors declare that there are no conflicts of interest.

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